



# Response of thicket swamp species to soil moisture levels: Implications for restoration

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## ABSTRACT

Wetlands dominated by shrubs are ubiquitous features in the Great Lakes Region of North America, yet little is known about the interactions between plants and the environmental conditions of these ecosystems. We measured the response of seven common wetland shrub species and two forbs to soil moisture treatments over a growing season. Average soil moisture content across three treatments ranged from < 30% to > 75%. These soil moisture differences produced a measurable effect on plant growth for six of the nine species tested. While *Rosa palustris* and *Spiraea alba* preferred the highest soil moisture treatments, *Salix interior*, *Sambucus nigra*, and *Solidago uliginosa* had higher biomass from moderate soil moisture, with no effect observed for *Cornus sericea*, *Salix discolor*, and *Viburnum lentago*. Most of the growth differences were from aboveground portions of the plants. A significant difference in root-to-shoot ratio (R:S) was found for eight of the species, with contrasting directions of change between the soil moisture treatments. New growth was compared to mean stomatal conductivity to determine an estimated water use efficiency (WUE<sub>e</sub>), which demonstrated a monotonic relationship for six of the species across the measured soil moisture regime. Only the upland goldenrod, *Solidago altissima*, demonstrated the expected increase in WUE<sub>e</sub> with decreasing soil moisture, which suggests wetland shrubs are inefficient at low soil moisture. The results of this study provide quantitative data on the range of responses to soil moisture of common species of shrub wetlands, which can be used by wetland managers to improve restoration success.

## 1. Introduction

Wetlands are species rich ecosystems characterized by soil moisture high enough to promote the formation of hydric soils, and are dominated by hydrophytic vegetation (National Wetlands Working Group, 1997). Wetlands perform a number of ecosystem services including flood mitigation, aquifer recharge, improved water quality, high species diversity, and habitat for regionally rare flora and fauna (Ballantine et al., 2012; Mitsch and Gosselink, 2015). Because of high ecological and social values including crucial habitat, floodwater protection, and nutrient and carbon storage, they are key targets of preservation and restoration across the globe (Bedford, 1999; Zedler, 2000; Brinson and Malvarez, 2002). Wetlands dominated by shrubs are known as thicket swamps or shrub carrs and are common landscape features of the Great Lakes Region of North America (National Wetlands Working Group, 1997; Johnston, 2003). Thicket swamps are species-diverse ecosystems containing shrubs less than 3 m tall and typically represent stable communities for many decades (Niering et al., 1986; Keith and Myerscough, 2003; Mensing et al., 1998). While experimental and field-

based correlative research on the environmental controls of ground-water level, light, and nutrient status on wetland assemblages including shrub species has been conducted (Jensen and Meyer, 2001; Kotowski et al., 2001; Duval et al., 2012), currently little is known about the response to soil moisture stress of species that dominate thicket swamps.

There is not a consensus on the role of shrubs in wetland ecosystems. Shrub encroachment into marsh and meadow systems are typically viewed as detrimental (Hanowski et al., 1999; Middleton, 2002a), leading to a reduction in species diversity and ecosystem services (Quintana-Ascencio et al., 2013; Budny and Benscoter, 2016). On the other hand, thicket swamps are key components of the fringing laggs of bogs, representing biodiversity hotspots and aiding in the maintenance of elevated bog water levels (Howie and van Meerveld, 2016). Thicket swamp species are also effective in an engineering context, preventing or delaying reforestation along electrical power-line corridors (Meilleur et al., 1994). In the Great Lakes Region of North America shrub thicket swamps are essential habitat for anuran and avian species, many of which are regionally threatened (Havens et al., 2002; Balcombe et al.,

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2005; Milne and Bennett, 2007; Eggers and Reed, 2015). Thicket swamps in the USA Great Lakes states have also been shown to have higher vascular plant species diversity than marshes (Stapanian et al., 2016; Stapanian et al., 2018). Regardless of the view of wetland shrubs as beneficial or detrimental to the landscape, a more thorough understanding of their tolerances to soil moisture would benefit management efforts of these species. Thicket species' responses to moisture levels will aid planning for habitat maintenance and creation for amphibian and bird species, particularly with increasing land use and climate change, or else providing insight into limits of range expansion for preservation of existing marsh or upland forest communities.

Previous research on temperate wetland plant response to moisture regimes has focussed on herbaceous and graminoid species. For example, soil saturation was found to be a significant factor in determining aboveground biomass and root adaptations in graminoid species (Rubio et al., 1995). Soil flooding can lead to decreased vegetative growth and lower shoot survivorship for some wetland sedges, grasses, and rushes (Liefers and Shay, 1981; Kercher and Zedler, 2004), while other sedge and grass species are tolerant of seasonal fluctuations in soil moisture, including flooding (Ewing, 1996; Didiano et al., 2018). Overall, the role of wetland soil flooding on plant species performance has received much more attention (Lenssen et al., 1999; Visser et al., 2000; Shi et al., 2015) than the opposite form of soil moisture stress, despite predicted future drying in most wetland systems as a consequence of climate change (Weltzin et al., 2003; Whittington and Price, 2006). Wetland shrubs are adapted to flooded conditions, and the physiological and morphological tolerance of most thicket swamp species to lower soil moisture is currently unknown beyond general drought indices that group some species into a narrow range of tolerance (Niinemets and Valladares, 2006).

In contrast to wetland systems, soil moisture scarcity is the norm in arid and semi-arid regions, and shrubs and other plant growth forms have adapted their growth patterns to the characteristic pulses of moisture supplied by irregular precipitation events (Schwinning and Sala, 2004). Arid system grasses scavenge water in the near-surface soil, while shrubs adopt a water conservation strategy (Golluscio and Oesterheld, 2007), increasing water use efficiency (WUE) during rooting-zone moisture stress (Letts et al., 2010). Dryland shrubs are relatively unresponsive to surface soil moisture stress and pulse rainfall events, utilizing extensive rooting systems to access deep moisture reserves (Hoover et al., 2017). Many wetland shrubs have shallow root architecture (Wallén, 1986), and thus may not be able to access a deeper water store in response to surface drying. This is particularly relevant in the context of climate change, where temperate region growing season precipitation regimes are predicted to shift to fewer storms of greater magnitude interspersed with longer dry periods (Sillmann et al., 2013; Wang et al., 2014). This rainfall pattern may increase rooting-zone moisture stress in temperate thicket swamps, with currently unknown effects on plant species growth.

The objectives of this study are to assess the response of several wetland shrub species common to temperate North America to three soil moisture regimes along an increasing wetness gradient. Specifically, this study addresses moisture tolerances of these species with respect to stem height, above- and belowground biomass, and the ratio between these. We also relate measured stomatal conductivity to net new growth to estimate the species' water use efficiency of growth in response to the imposed moisture gradient. The results of this study provide much needed information on the relationship between wetland shrubs and water availability, and suggest that dominant thicket swamp species exhibit different responses to soil moisture. These documented moisture tolerances can be used in thicket swamp restoration projects, prediction of shrub encroachment to sedge meadows, and land management in the face of changes to the precipitation regime.

**Table 1**  
Species characteristics of the study.

| Species                                                       | Family         | Growth Form | Common name       |
|---------------------------------------------------------------|----------------|-------------|-------------------|
| <i>Cornus sericea</i> L. ssp. <i>sericea</i>                  | Cornaceae      | Shrub       | Red osier dogwood |
| <i>Rosa palustris</i> Marshall                                | Rosaceae       | Shrub       | Swamp rose        |
| <i>Salix discolor</i> Muhl.                                   | Salicaceae     | Shrub       | Pussy willow      |
| <i>Salix interior</i> Rowlee                                  | Salicaceae     | Shrub       | Sandbar willow    |
| <i>Sambucus nigra</i> L. ssp. <i>canadensis</i> (L.) R. Bolli | Caprifoliaceae | Shrub       | Common elderberry |
| <i>Spiraea alba</i>                                           | Rosaceae       | Shrub       | White meadowsweet |
| <i>Viburnum lentago</i> L.                                    | Caprifoliaceae | Shrub       | Nannyberry        |
| <i>Solidago altissima</i> L.                                  | Asteraceae     | Forb        | Canada goldenrod  |
| <i>Solidago uliginosa</i> Nutt.                               | Asteraceae     | Forb        | Bog goldenrod     |

## 2. Materials and methods

### 2.1. Study species and common garden experiment

We selected seven native shrub species common to eastern North American thicket swamps (Table 1). The species tested ranged through four families, were all perennial, and were selected to represent a range of observed tolerances to wet conditions. We also tested two herbaceous goldenrods, one typical of thicket swamps, the other the dominant species at the experimental field site. We did not include the common *Alnus incana* ssp. *rugosa* in our experiment due to difficulty finding donor sites with juvenile specimens. All shrubs and *Solidago uliginosa* were grown in common conditions in isolated containers within a wet meadow for one year prior to experiment initiation. The plant specimens were removed from their potting soil, rinsed of residual soil, and weighed immediately prior to planting for this experiment.

A common garden experiment was initiated in mid-May in a 2.5 ha field (43° 40' 49" N, 79° 50' 01" W) that had been in agricultural use for the previous eight decades but had been left fallow for the immediate five years prior to the study. This site was slated to be restored back to the original thicket swamp community as part of compensation for surrounding residential subdivision construction, and this project was designed to assess the potential of the site for wetland restoration. The fallow field was vegetated by various obligate and facultative upland herbaceous species, dominated by *Solidago altissima*, *Solidago canadensis*, *Aster vimineus*, *Euthamia graminifolia*, *Sonchus arvensis*, and *Geum aleppicum*. The site was located within the Peel Plains physiographic region and soils were silty-clay loams with 30–40 cm of organic-enriched topsoil. Soil porosity in the topsoil averaged 68%. The water table of the site fluctuated between 0 and 150 cm below ground, and averaged 52 cm below ground surface during the study period.

We tested the effect of soil moisture on wetland shrub growth through three watering treatments: Control, Moist, and Wet. The Control treatments were subject to ambient precipitation during the experiment. To increase the soil moisture in the Moist and Wet treatments a 6 mil reinforced polyethylene barrier lined the 30-cm diameter, 60-cm deep planting location of each replicate (Fig. S1). The barrier prevented drainage of ambient rainwater away from the root zone of each plant. Rain water was collected at the site and applied to the Wet treatment plants such that ambient precipitation was doubled. Each of the nine species were replicated nine times for each treatment, for a total of 243 shrubs and forbs. Replicates were planted 1 m apart from each other in a cleared portion of the fallow field. The experiment lasted 15 weeks, until the very end of August, whereupon plants were harvested.

### 2.2. Measurements and statistical analysis

Soil moisture in the rooting zone of each plant was measured weekly with a Hydrosense CS620 soil measurement system (Campbell

Scientific, Utah, USA) with 20-cm probes. Stomatal conductivity was measured weekly on three leaves of each plant between 10:00 and 14:00 with an AP-4 porometer (Decagon, UK) calibrated at least every 90 min during measurement periods. Values were only included when photosynthetically active radiation was greater than  $500 \mu\text{mol m}^{-2} \text{s}^{-1}$ , as determined by an SQ-222 quantum sensor (Apogee, Utah, USA) connected to an on-site weather station. Air temperature and rainfall were measured on site at 2 m above ground with an HMP60 temperature and relative humidity sensor (Vaisala, Finland) and R.M. Young tipping-bucket rain gauge, respectively, connected to a CR1000 data logger (Campbell Scientific, Utah, USA).

Biomass and stem height were used to measure plant performance in response to soil moisture variability. Initial fresh mass for each replicate was determined upon removal from the wet meadow donor site and rinsing and drip-drying of the roots. At the end of the 15-week experimental period plants were harvested and roots were rinsed clean of soil. Plants were allowed to drip dry and then were weighed for fresh mass. All biomass was then dried at  $40^\circ\text{C}$  until constant mass and re-weighed for dry mass. The initial fresh mass of each replicate was subtracted from the fresh mass after the 15 weeks to determine the new growth fresh mass. New growth dry mass was then estimated through the calculated ratio of final fresh to dry mass. Stem heights were measured every three weeks during the experiment; however, analysis of within-season variability yielded no clear differences between treatments, and only the comparison between initial and final stem heights were included in the analysis. To explain variability in plant biomass due to soil moisture we calculated the water use efficiency (WUE) of each plant using the new growth relative to stomatal conductance. Because we could not measure whole-plant transpiration or the instantaneous photosynthetic rate, this approach provided only an estimate of WUE, which we denote  $\text{WUE}_e$ .

The effect of the soil moisture treatments on plant performance (proportional increase in stem height, final above- and belowground biomass, and root-to-shoot ratio) was analysed with one-way ANOVA for each species. Data were checked for normality and homogeneity using the Levene and Shapiro-Wilk tests, respectively. Significant differences between the treatments were assessed with Tukey's post hoc test. Effect sizes between treatments were determined using Cohen's  $d$ . Linear and nonlinear regressions were performed between average soil moisture of each plot and  $\text{WUE}_e$ . All analyses were performed in Statistica 8. All biomass data represent dry-weight basis.

### 3. Results

#### 3.1. Weather and soil moisture variation

The atmospheric conditions of the field experiment were representative of the long-term mean conditions for the study area (Table 2). The month of May was nearly  $3^\circ\text{C}$  warmer than average, while the other months were very close to average. Overall, the experiment was conducted during a period that was 24% wetter than

**Table 2**

Seasonal weather variation during the experiment. Temperature values are averages and precipitation values are totals  $\pm$  standard deviation for the most recent climate normal period (1981–2010) from Toronto Lester B. Pearson International Airport (WMO ID# 71624), located  $< 15$  km from the study site.

| Month  | Temperature ( $^\circ\text{C}$ ) |                | Precipitation (mm) |                  |
|--------|----------------------------------|----------------|--------------------|------------------|
|        | Study season                     | Climate normal | Study season       | Climate normal   |
| May    | 16.3                             | $13.1 \pm 1.9$ | 55.0               | $74.3 \pm 27.8$  |
| June   | 18.9                             | $18.6 \pm 1.6$ | 100.6              | $71.3 \pm 40.4$  |
| July   | 22.3                             | $21.5 \pm 1.5$ | 181.8              | $75.7 \pm 38.8$  |
| August | 20.9                             | $20.6 \pm 1.5$ | 33.6               | $78.1 \pm 44.6$  |
| Season | 19.3                             | $18.5 \pm 1.7$ | 371.0              | $299.5 \pm 71.3$ |

**Table 3**

Variation in soil moisture between the three treatments. Values are means  $\pm$  standard deviation of nine replicates measured weekly for each of the Control, Moist, and Wet treatments for each species. For each species, there were significant differences ( $P < .01$ ) between all soil moisture treatments as determined by one-way ANOVA.

| Species                   | Volumetric soil moisture (%) |                |                |
|---------------------------|------------------------------|----------------|----------------|
|                           | Control                      | Moist          | Wet            |
| <i>Cornus sericea</i>     | $35.9 \pm 5.4$               | $52.0 \pm 6.7$ | $69.9 \pm 4.6$ |
| <i>Rosa palustris</i>     | $39.3 \pm 4.0$               | $51.1 \pm 4.6$ | $66.4 \pm 9.1$ |
| <i>Salix discolor</i>     | $35.4 \pm 2.6$               | $47.0 \pm 5.7$ | $70.0 \pm 7.0$ |
| <i>Salix interior</i>     | $38.4 \pm 2.0$               | $51.4 \pm 4.4$ | $69.0 \pm 8.6$ |
| <i>Sambucus nigra</i>     | $40.7 \pm 4.7$               | $57.2 \pm 4.6$ | $74.5 \pm 6.3$ |
| <i>Spiraea alba</i>       | $38.4 \pm 2.0$               | $51.4 \pm 4.4$ | $69.0 \pm 8.6$ |
| <i>Viburnum lentago</i>   | $26.9 \pm 2.7$               | $36.1 \pm 3.5$ | $48.0 \pm 3.7$ |
| <i>Solidago altissima</i> | $39.3 \pm 4.8$               | $49.2 \pm 0.6$ | $60.0 \pm 7.0$ |
| <i>Solidago uliginosa</i> | $41.9 \pm 2.8$               | $63.0 \pm 4.5$ | $78.7 \pm 4.9$ |

normal, ranking in the upper 15% of seasonal precipitation total from the climate normal. Rainfall in May was 35% lower than the long-term average, though still within one standard deviation. August of the study period was also much drier than average, drier than all but three years of the most recent climate normal. In contrast, June and July were 41 and 140% wetter than normal, respectively, with July being wetter than all but one year of the current climate normal. In particular, the elevated rain in July was mostly due to a 126-mm event that occurred over a few hours on 08-July.

The experimental conditions resulted in significant moisture gradients between the treatments for all the wetland shrubs tested (Table 3). Mean VMC of the Control treatments ranged from a low of  $26.9 \pm 2.7\%$  for *Viburnum lentago* to a high of  $41.9 \pm 2.7\%$  for *Solidago uliginosa*. These two species represented the lowest and highest mean VMC for the Moist and Wet treatments as well. VMC declined throughout the growing season, from highs of  $\sim 70\%$  (representing saturation) for the Wet treatment of *Salix discolor* to end-of-season lows of  $\sim 20\%$  for the Control treatment of most species. This seasonal decrease was interrupted as a result of rain events, most notably the storm in early July where VMC increased on average by an absolute 10% for the Control treatments and upwards of 30% for the Wet treatment. Generally, there was an absolute increase of 10–15% in seasonally-averaged VMC between Control and Moist treatments, and an equivalent increase between Moist and Wet treatments. The minimum difference ( $\sim 20\%$ ) between the Control and the Wet treatments occurred in the *Solidago altissima* plants, whereas there were absolute differences between Control and Wet treatments of greater than 33% for *Salix discolor*, *Sambucus nigra*, and *S. uliginosa* plants.

#### 3.2. Wetland shrub growth

The differences in rooting-zone soil moisture between the three treatments elicited growth responses for most of the wetland shrubs tested, in addition to the two forbs. There were significant differences in new stem growth (measured as increase in height) between the soil moisture treatments for four of the seven wetland shrubs tested (Table 4). There was a ladder-like response in stem growth to soil moisture for *Spiraea alba*, with the Wet treatment resulting in nearly twice as much stem growth as the Control (Fig. 1a; S2). Two of the shrubs with differences in stem growth had a plateau response to soil moisture: new stem growth in *Rosa palustris* and *Salix interior* was restricted in the field Control, with between 50 and 125% greater growth in the two wetter treatments. *Sambucus nigra* stem growth peaked in the Moist treatment, with average new growth 50 and 100% greater than the drier and wetter treatments, respectively. Stem growth responded positively to increased soil moisture, with a doubling in new growth between the Control and Wet treatments. New stem growth differed

**Table 4**

Results of ANOVA on new stem height growth in response to soil moisture treatments. There were two degrees of freedom between treatments for each species.

| Species                   | F     | P-value  | Effect sizes between treatments |            |             |
|---------------------------|-------|----------|---------------------------------|------------|-------------|
|                           |       |          | Cont – moist                    | Cont – wet | Moist – wet |
| <i>Spiraea alba</i>       | 4.76  | 0.019    | 0.68                            | 1.60**     | 0.77        |
| <i>Rosa palustris</i>     | 5.24  | 0.014    | 1.58**                          | 1.09*      | 0.48        |
| <i>Salix interior</i>     | 13.62 | 0.0001   | 2.25***                         | 2.25***    | 0.36        |
| <i>Sambucus nigra</i>     | 16.50 | < 0.0001 | 1.93***                         | 0.80       | 2.73***     |
| <i>Cornus sericea</i>     | 1.81  | 0.186    | 0.83                            | 0.21       | 0.84        |
| <i>Salix discolor</i>     | 2.03  | 0.154    | 0.95                            | 0.58       | 0.37        |
| <i>Viburnum lentago</i>   | 1.06  | 0.363    | 0.39                            | 0.36       | 0.63        |
| <i>Solidago altissima</i> | 4.93  | 0.016    | 1.34*                           | 1.22*      | 0.09        |
| <i>Solidago uliginosa</i> | 5.97  | 0.008    | 1.46**                          | 0.59       | 1.03*       |

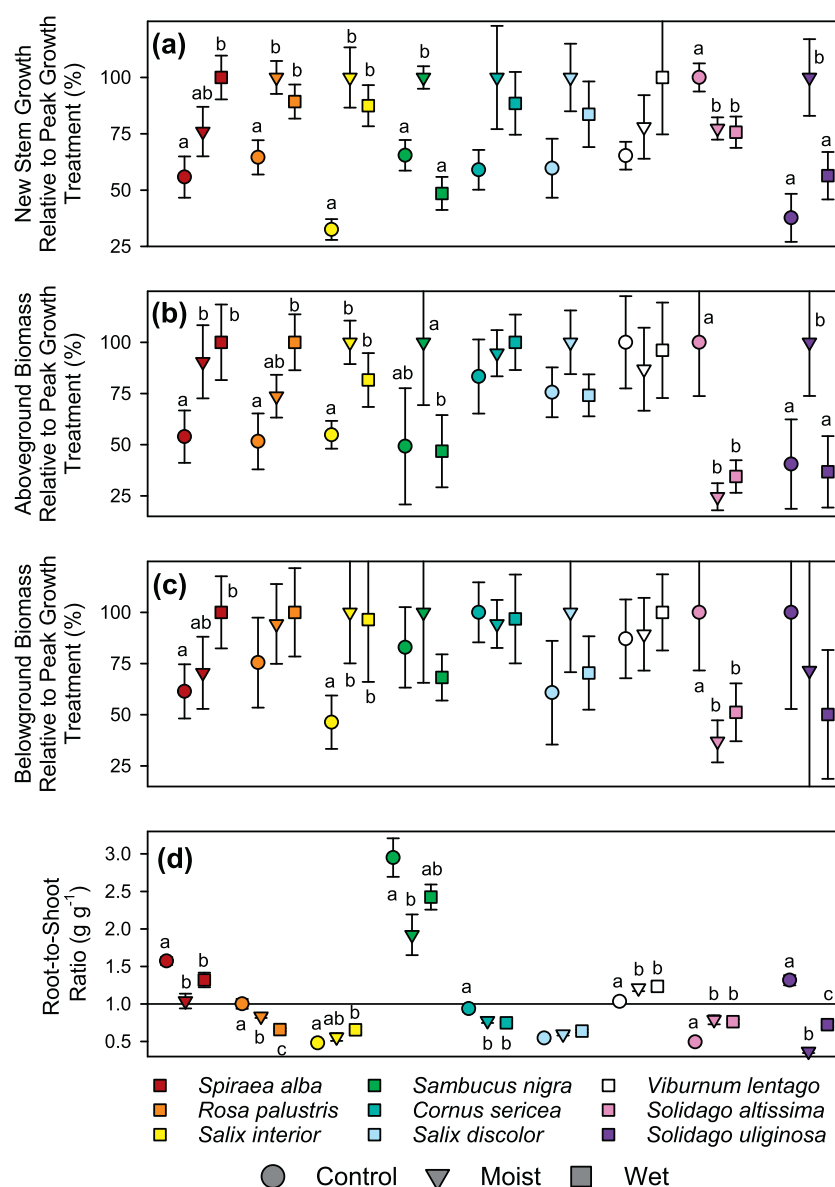
\* $P < .05$ ; \*\*  $P < .01$ ; \*\*\*  $P < .001$ .

**Table 5**

Results of ANOVA on aboveground biomass in response to soil moisture treatments. There were two degrees of freedom between treatments for each species.

| Species                   | F     | P-value | Effect sizes between treatments |            |             |
|---------------------------|-------|---------|---------------------------------|------------|-------------|
|                           |       |         | Cont – moist                    | Cont – wet | Moist – wet |
| <i>Spiraea alba</i>       | 20.23 | < 0.001 | 2.18***                         | 2.42***    | 0.77        |
| <i>Rosa palustris</i>     | 10.07 | < 0.001 | 0.98                            | 1.67**     | 0.97        |
| <i>Salix interior</i>     | 20.06 | < 0.001 | 3.41***                         | 1.91***    | 1.00        |
| <i>Sambucus nigra</i>     | 4.78  | 0.018   | 0.99                            | 0.10       | 1.15*       |
| <i>Cornus sericea</i>     | 1.43  | 0.260   | 0.55                            | 0.73       | 0.26        |
| <i>Salix discolor</i>     | 2.95  | 0.071   | 0.94                            | 0.06       | 1.01        |
| <i>Viburnum lentago</i>   | 0.98  | 0.388   | 0.60                            | 0.17       | 0.55        |
| <i>Solidago altissima</i> | 13.81 | < 0.001 | 1.88**                          | 1.66**     | 0.92        |
| <i>Solidago uliginosa</i> | 6.28  | 0.006   | 1.23*                           | 0.12       | 1.38**      |

\* $P < .05$ ; \*\*  $P < .01$ ; \*\*\*  $P < .001$ .



**Fig. 1.** Growth response of the nine species tested for the three soil moisture treatments depicting: (a) stem height response; (b) aboveground biomass response; (c) belowground biomass response, and (d) the root-to-shoot ratio in response to the soil moisture treatments. Data for (a)–(c) represent response of each species to the soil moisture treatments relative to the treatment (Control, Moist, Wet) for each species that produced the greatest growth, which was standardized to 100%. Values are mean (+SE) of nine replicates per treatment. Different letters within each graph indicate significant differences between soil moisture treatments ( $p < .05$ ).



between soil moisture treatments for both goldenrods tested, with *Solidago altissima* under the Control treatment outperforming the wetter treatments by 30%, while *Solidago uliginosa* grew more in the Moist treatment (Fig. 1a).

The final aboveground biomass response to the soil moisture treatments was similar to stem growth, with four of seven shrubs and both goldenrods being affected (Table 5). *Salix interior* and *Spiraea alba* aboveground biomass was strongly restricted in the drier soils, with highly significant differences and effect sizes close to or greater than 2.0 between the two wetter treatments and the Control. Aboveground biomass of *Rosa palustris* exhibited the ladder-like increase in response to increased soil moisture, with Wet treatment mean biomass nearly twice the aboveground biomass of the Control treatment (Fig. 1b; S3). In contrast, *Sambucus nigra* aboveground biomass was lowest in the Wet soil moisture treatment, which was less than half the aboveground biomass of the Moist treatment. This response pattern was also found for the wetland goldenrod, *Solidago uliginosa*, where the Moist treatment elicited more than twice the aboveground biomass than either wetter or drier soil treatments. The facultative upland goldenrod, *Solidago altissima*, performed poorly in the wetter treatments, with average aboveground biomass nearly three times higher in the Control treatment (Fig. 1b). Changing soil moisture did not have an effect on aboveground biomass for either *Cornus sericea* or *Viburnum lentago*.

The response of belowground biomass to soil moisture regime was less significant than for the aboveground response, with only two shrubs and one forb being affected (Table 6). Mean belowground biomass for *Spiraea alba* was ~60% greater in the Wet treatment than the Control (Fig. 1c; S3). Mean root mass of *Salix interior* was more than two-times greater in the Moist and Wet treatments than the Control treatment. As with the aboveground biomass, the Control treatment belowground biomass for *Solidago altissima* was approximately two-times greater than the Moist and Wet treatments.

The differences, and lack of difference, between soil moisture treatments in above- and belowground biomass resulted in significant differences in the root-to-shoot ratio between soil moisture treatments for eight of the nine species tested (Table 7). Only the ratio between above- and belowground biomass for *Salix discolor* was unaffected by soil moisture. The trend in changing root-to-shoot ratio due to soil moisture treatment varied between the species (Fig. 1d). Increasing soil moisture led to a significant decrease in root-to-shoot ratio for *Rosa palustris*. This trend was also evident in *Cornus sericea*, despite no significant differences in above- or belowground biomass between treatments. The opposite trend, increasing ratio with increasing soil moisture, occurred in *Salix interior*, *Solidago altissima*, and *Viburnum lentago*, the latter also not having significant differences in biomass values. A third trend, with the Moist treatment producing greater relative aboveground growth leading to a low root-to-shoot ratio, was found for *Solidago uliginosa*. This trend appeared to also occur for

**Table 6**

Results of ANOVA on belowground biomass in response to soil moisture treatments. There were two degrees of freedom between treatments for each species.

| Species                   | F    | P-value | Effect sizes between treatments |            |             |
|---------------------------|------|---------|---------------------------------|------------|-------------|
|                           |      |         | Cont – Moist                    | Cont – Wet | Moist – Wet |
| <i>Spiraea alba</i>       | 4.49 | 0.022   | 0.40                            | 1.31*      | 0.92        |
| <i>Rosa palustris</i>     | 2.78 | 0.082   | 0.94                            | 0.98       | 0.24        |
| <i>Salix interior</i>     | 7.45 | 0.003   | 1.73**                          | 1.88**     | 0.10        |
| <i>Sambucus nigra</i>     | 2.24 | 0.128   | 0.48                            | 0.47       | 1.10        |
| <i>Cornus sericea</i>     | 0.30 | 0.742   | 0.45                            | 0.19       | 0.15        |
| <i>Salix discolor</i>     | 3.32 | 0.053   | 1.04                            | 0.34       | 0.90        |
| <i>Viburnum lentago</i>   | 1.80 | 0.186   | 0.13                            | 0.80       | 0.76        |
| <i>Solidago altissima</i> | 5.79 | 0.009   | 1.28*                           | 1.00*      | 0.80        |
| <i>Solidago uliginosa</i> | 1.60 | 0.222   | 0.43                            | 0.81       | 0.42        |

\*  $P < .05$ ; \*\*  $P < .01$ ; \*\*\*  $P < .001$ .

**Table 7**

Results of ANOVA on the root-to-shoot biomass ratio in response to soil moisture treatments. There were two degrees of freedom between treatments for each species.

| Species                   | F      | P-value | Effect sizes between treatments |            |             |
|---------------------------|--------|---------|---------------------------------|------------|-------------|
|                           |        |         | Cont – Moist                    | Cont – Wet | Moist – Wet |
| <i>Cornus sericea</i>     | 7.74   | 0.003   | 1.25*                           | 1.51**     | 0.39        |
| <i>Rosa palustris</i>     | 17.39  | < 0.001 | 1.14*                           | 2.39***    | 2.86***     |
| <i>Salix discolor</i>     | 3.16   | 0.061   | 0.61                            | 1.06       | 0.70        |
| <i>Salix interior</i>     | 5.06   | 0.015   | 0.72                            | 1.86**     | 0.68        |
| <i>Sambucus nigra</i>     | 3.82   | 0.036   | 1.30*                           | 0.81       | 0.74        |
| <i>Spiraea alba</i>       | 9.79   | < 0.001 | 2.26***                         | 1.08*      | 0.96        |
| <i>Viburnum lentago</i>   | 6.14   | < 0.001 | 1.60**                          | 1.46**     | 0.15        |
| <i>Solidago altissima</i> | 13.34  | < 0.001 | 2.12***                         | 3.24***    | 0.18        |
| <i>Solidago uliginosa</i> | 110.95 | < 0.001 | 6.27***                         | 3.69***    | 4.19***     |

\*  $P < .05$ ; \*\*  $P < .01$ ; \*\*\*  $P < .001$ .

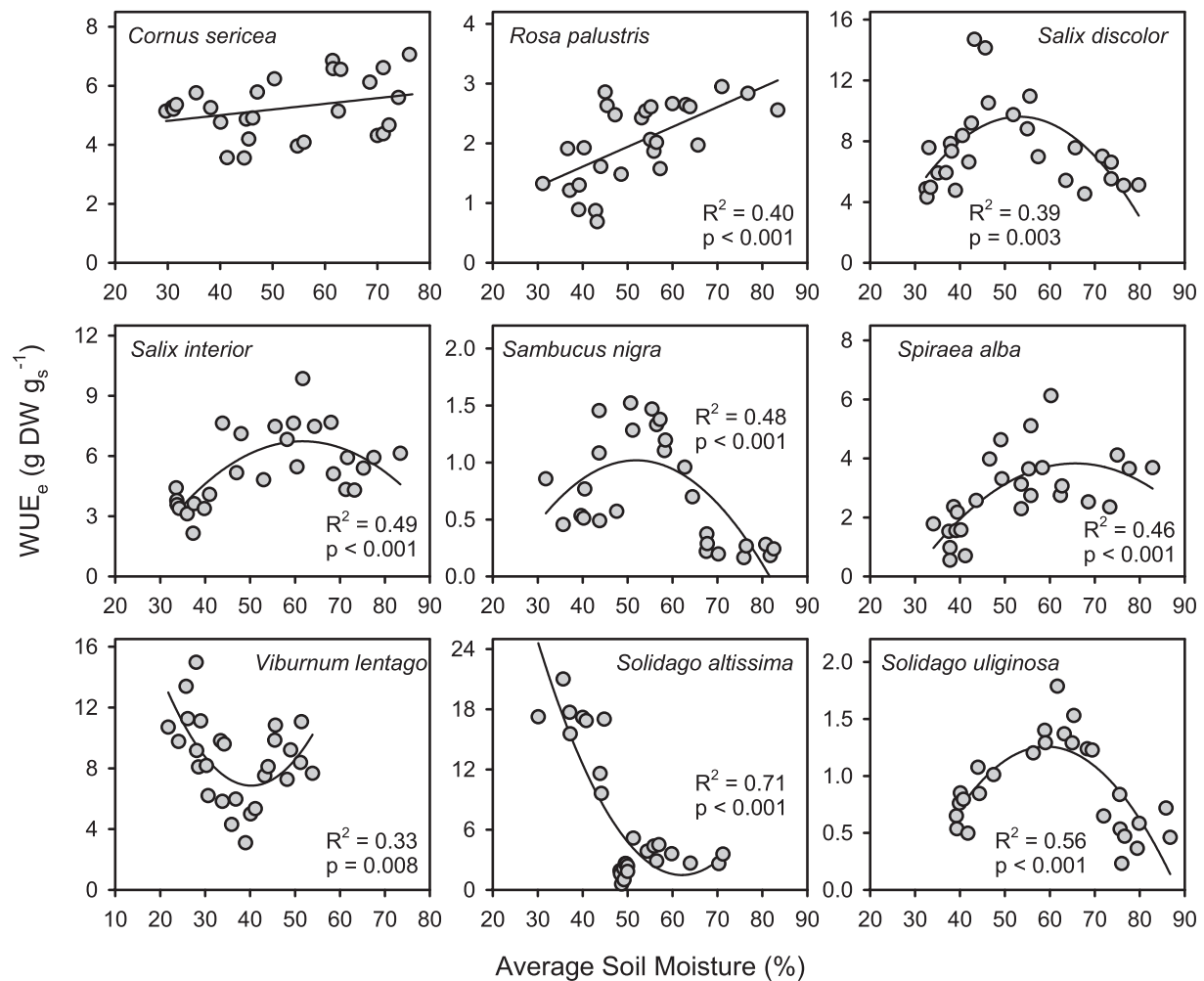
*Sambucus nigra* and *Spiraea alba* (Fig. 1d), though statistically there were no differences in root-to-shoot ratio between the Moist and Wet treatments.

Root-to-shoot ratio was less than 1.0 for all three soil moisture treatments for three shrubs and one forb, and much greater than 1.0 for all three treatments for *Sambucus nigra* (Fig. 1d). The differences in soil moisture between the three treatments also led to sizable shifts in root-to-shoot ratios for some species. Mean root-to-shoot ratio for *Solidago uliginosa* dropped from 1.32 in the Control to 0.72 and 0.37 in the Wet and Moist treatments, respectively. Similar decreases were found for *Cornus sericea* and *Rosa palustris*, where root-to-shoot ratios near 1.0 for the Control decreased with increasing soil moisture, to 0.75 and 0.66 for the Wet treatments, respectively. In contrast, the increased ratio associated with *Viburnum lentago* was from a range of values above and below 1.0 (mean of 1.03) in the Control to eight of nine replicates greater than 1.1 (mean of 1.24) in the Wet treatment. Overall, the F-scores, levels of significance, and effect sizes for the differences between soil moisture treatments were higher for the root-to-shoot ratio (Table 7) than either above- or belowground biomass separately (Tables 5 & 6).

### 3.3. Water resource use

The differences in biomass growth in response to soil moisture differences are in part a result of the efficiency with which the species utilize the available water. Fig. 2 depicts the individual replicates' estimated water use efficiency ( $WUE_e$ ) for each species as a function of mean soil moisture. A significant relationship was found for all species except *Cornus sericea*. The relationship between average soil moisture and  $WUE_e$  followed four different trends. The most common response was a quadratic relationship with peak  $WUE_e$  at intermediate VMC. This response pattern was found for four shrubs and one forb (Fig. 2). The predicted  $WUE_e$  peaked at different mean soil moistures, ranging from 54% for *Sambucus nigra* to 66% for *Spiraea alba* for the shrubs, and 59% for the forb *Solidago uliginosa*. These differing soil moistures represent the conditions at which each species can most efficiently maximize new biomass for minimum rates of stomatal conductance. The results demonstrate most temperate wetland shrubs are most efficient at moderate soil moistures.

On the other hand, the remaining species did not have peak  $WUE_e$  occur at intermediary average soil moistures (Fig. 2). *Rosa palustris* replicates continued to increase  $WUE_e$  with increasing soil moisture, nearly tripling in efficiency across the range of measured soil moisture. Though not significant, the general trend for *Cornus sericea* also followed this relationship. In contrast, there was more than a three-fold decrease in  $WUE_e$  with increasing soil moisture for the facultative upland forb, *Solidago altissima*. Average soil moisture, and the range between treatments, was lower for *Viburnum lentago* plots than the other



**Fig. 2.** Estimated water use efficiency of the tested plants in response to seasonally-averaged soil moisture in each replicate. Regression correlations and significance of the relationship for each species tested are indicated.

species; however, the plants still responded to this narrower range, with a predicted minimum  $WUE_e$  at 40% VMC and increased efficiencies with increasing and decreasing soil moisture.

#### 4. Discussion

Studies of water-stress impacts on plant growth are focused on dryland species (Galmés et al., 2005; Evans et al., 2013; McAbee et al., 2017), with much less research on wetland environments. This is the first study to quantitatively assess the response of temperate thicket swamp species to soil moisture. The seven shrubs tested demonstrate a range of growth responses to soil moisture, with species being insensitive to soil moisture (three species tested), increasing growth with increasing soil moisture (three species), or finding optimum growth at intermediary soil moisture levels (one species). The latter response was also found for the wetland forb tested, while the facultative upland forb performed poorly in wetter soil conditions. These responses were due in part to differences in stomatal conductance, which affected species' water use efficiencies across a wetness gradient.

##### 4.1. Aboveground biomass more responsive to soil moisture than belowground biomass

Results of this study demonstrate that over one growing season most of the response to soil moisture differences in wetland plants will be manifest in the aboveground portion (Tables 4–5), with limited

expression in root biomass (Table 6). In particular, the new stem growth in the Wet treatment was twice the new growth in the Control treatment for *Spiraea alba* and *Salix interior* (Fig. 1a), clearly demonstrating a preference of those species for soils near saturation (Table 2). In contrast, stem elongation of *Sambucus nigra* and *Solidago uliginosa* in the Moist treatment was between 50 and 100% greater than the Control or Wet treatments (which corresponded with biomass differences), reflecting those species' preferences for moderate soil moisture. Results from other species, such as *Viburnum lentago* and *Salix discolor*, suggest soil moisture may have an effect on growth, but not over the timescale of this experiment. It is possible that the significant differences observed above reflect species that are more sensitive than others, such that the changes were expressed over a single growing season. Galmés et al. (2005) have shown that upland shrubs grow more slowly in general, and in response to water stress in particular, than other growth forms, which may hold true for some wetland shrubs. Over a longer experimental period the other species may have developed differential responses to soil moisture regime, particularly in the belowground portion (Schultz and Pett, 2018).

The differential response between above- and belowground portions led to significant differences in the root-to-shoot ratio for eight of the species tested (Table 7; Fig. 1d). The decreased ratio for three shrubs and one goldenrod as soil moisture increased was presumably due to alleviation of a moisture stress allowing for increased partitioning of assimilated carbon to shoots and leaves. It is somewhat surprising then that the root-to-shoot ratio increased with increasing moisture for the

shrubs *Salix interior* and *Viburnum lentago*, and the upland goldenrod *Solidago altissima*. Root extension in response to flooded soil conditions has been observed for wetland species in order to maintain root-zone aerobic conditions (Visser et al., 2000; Kercher and Zedler, 2004). McAbee et al. (2017) suggest water regime can dictate partitioning of synthesized carbon to belowground biomass of cold-desert shrubs. Given that both shoots and roots of *S. interior* were negatively affected by the drier Control treatment it is possible that soil moisture of that treatment was prohibitive to growth such that our observed reduction in root-to-shoot ratio was an early indication of that species' inability to survive in that moisture regime. Additionally, without significant differences in stem height, above-, or belowground biomass between soil moisture treatments for *V. lentago* it is difficult to speculate on the cause of increased root-to-shoot ratio with increased moisture. Further research is warranted on the apportionment of new growth to the various components of wetland shrubs.

#### 4.2. Contrasting utilization of water resources across a moisture gradient

As soil moisture becomes limiting the general terrestrial plant response is to decrease stomatal conductance to reduce water loss (Reed and Loik, 2016). The reduction in transpiration rate in response to soil moisture stress typically leads to an increase in WUE, as production is affected less (Zhang et al., 2018). The results of this study are largely at odds with this general trend, with only the terrestrial forb *Solidago altissima* demonstrating this water-carbon response to decreasing soil moisture. We note that our values of estimated water use efficiency (WUE<sub>e</sub>) are not directly comparable with the literature as we were unable to measure instantaneous rates of photosynthesis; however, the general trends are informative. The unimodal response in WUE<sub>e</sub> to soil moisture found in four of the wetland shrubs tested, and the wetland goldenrod (Fig. 2), is in part due to an inability to reduce stomatal conductivity at lower soil moisture levels. The apparent lack of water conservation for *Cornus sericea*, *Salix discolor*, *Spiraea alba*, and *Solidago uliginosa* (Table S1; Fig. S4) suggests that thicket swamp species have limited ability to reduce water loss through stomatal closure under conditions considered dry for a thicket swamp. Stanton and Mickelbart (2014) did observe significantly lower stomatal conductivity for *S. alba* plants under moisture stress, suggesting wetland shrubs do have some stomatal control in extreme dry conditions. That finding was in response to a six-week drought period, however, which is much longer than the plants experienced in the present study, and a period not normally found in wetland environments.

There is very little research on plant-water relations and stomatal control for thicket swamp species, particularly in response to changes in soil moisture. Lafleur (1988) found mature *Salix discolor* shrubs have lower stomatal conductance than a dominant sedge in a subarctic marsh, though the data were not related to soil moisture. The data available on stomatal control of other wetland plant growth forms in response to soil moisture supports the idea that wetland plants in general do not tightly regulate transpiration. Svejcar and Riegel (1998) found no relationship between graminoid stomatal conductance and water level in alpine meadows. Lower water level had no effect on stomatal conductivity of a common marsh grass, but led to greater biomass production (Glaeser et al., 2016), implying a higher WUE under drier conditions. The majority of woody species tested on tree islands of the Pantanal wetland by Dalmagro et al. (2016) increased stomatal conductivity during the dry season relative to flooded conditions. In that environment flooding was the stress rather than drought, and led to lower rates of photosynthesis and WUE. In contrast, Xu et al. (2015) found reduced WUE for several alpine marsh grasses and sedges in response to low soil moisture, while flooding of soils led to lower WUE for forbs. It is possible that the very high rates of stomatal conductivity of wetland shrub species (Frédette et al., 2019) limits their ability to restrict water loss in times of moisture stress. The majority of the thicket swamp species tested here had lower WUE<sub>e</sub> at both high and

low moisture levels (Fig. 2), which was more related to growth (Fig. 1b), though four of the seven tested shrubs did have lower stomatal conductivity in Control relative to Wet plots (Fig. S4). While mean stomatal conductivity in the Control soil moisture treatment was significantly lower than in the Wet treatment for *Salix interior* and *Sambucus nigra*, this did not translate into a difference in WUE<sub>e</sub>. The evidence presented here suggests that while some vascular plant species typical of thicket swamps are able to reduce stomatal conductivity in line with lower soil moisture, this reduction in transpirative water loss is largely disconnected from plant growth. Elucidating the physiological mechanisms responsible for this disconnect would improve our understanding of wetland shrub growth.

#### 4.3. Implications for wetland management

The results of this study provide valuable information to land managers tasked with improving species diversity and other ecosystem services of natural, restored, and created wetlands. It is well known that wetland grasses, sedges, and forbs spatially segregate across natural sites in line with moisture tolerances (Moran et al., 2008; Duval et al., 2012; Hajek et al., 2013), and restoration efforts have used this knowledge in strategic planting designs (van der Valk et al., 1999; Duval et al., 2010; Ter Heerdt et al., 2017). With a thorough understanding of the hydrological regime of a site it is also possible to develop strategic planting plans that position certain shrub thicket species in appropriate zones based on their water requirements. This will lead to an overall increase in success of community maintenance and (re-) establishment. Specifically, *Spiraea alba* and *Rosa palustris* should be reserved for the wettest areas of a site and/or a wetland restoration/creation project known or expected to remain near saturation for a significant portion of the growing season. *Sambucus nigra* and the forb *Solidago uliginosa* prefer moderate moisture regimes, with limited success in both too dry and too wet environments. The two willow species tested appear to be more tolerant of drier conditions than the previous examples, and the evidence suggests they are tolerant of both moderate and higher soil moisture regimes.

Other thicket swamp species have much broader moisture tolerances. Of the species tested in this field experiment *Cornus sericea* and *Viburnum lentago* are most likely to grow across a range of soil moisture conditions, and are therefore suitable species for rapid establishment in most wetland restoration planting regimes. The moisture tolerance of *C. sericea* identified here may also partially explain the success of that species' range expansion into sedge meadows and fens (Middleton, 2002a, 2002b), which are typically drier than thicket swamps. On the other hand, significant die-off or reduced growth of existing resident plant populations of wetland restoration projects should be expected with the creation of wetter environments. This is exemplified by the one terrestrial species we tested, *Solidago altissima*, experiencing reduced stem elongation, lower aboveground and belowground biomass, and lower water use efficiency in the wetter soil moisture treatments.

The success of any restoration design in large part depends on the coupling of a site's soil moisture with suitable species adapted to that moisture regime (Bedford, 1999). The soil moisture regime is determined by the water balance of a site (Magee and Kentula, 2005); therefore, field knowledge or predictive modelling of the hydrological fluxes at a thicket swamp restoration or creation site will aid in species selection and placement (Large et al., 2007; Booth and Loheide II, 2012). The higher than normal rainfall during the growing season of the present experiment likely aided the species adapted to wetter conditions, especially in the Wet treatment. Because of the natural water balance at the site many introduced wetland shrub species would likely experience reduced growth in years characterized by average precipitation without the incorporation of added moisture from an external source. Over the long term, this may lead to the dominance of a few thicket species tolerant of a lower moisture regime, reducing overall plant and animal species richness and diversity at the site (Barbosa da



Silva et al., 2016; Robichaud and Rooney, 2017; Du et al., 2017). If a species-rich restored thicket swamp is the desired goal, land managers should ensure a greater flux of water is delivered to the site.

### Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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### Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.ecoleng.2020.105911>.

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